Advanced Causal Inference

Lecture 14: Instrumental Variable

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This lecture note is partially based on the lecture notes/textbook of Ying Zhou, Eric Tchetgen Tchetgen

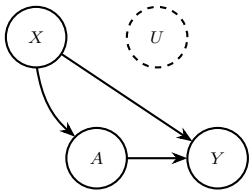
Outline

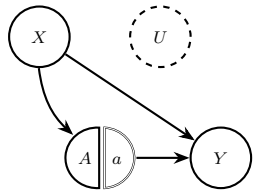
1. Overview
2. Review: Linear Regression
3. Instrumental Variable
4. LATE/CACE
5. ATT
6. ATE
7. TSLS
8. Nonparametric IV
9. Additional Topics
10. Assignment

Overview

- We have assumed no unmeasured confounding between $A - Y$

$$Y(a) \perp\!\!\!\perp A \,|\, X \;, \quad \forall a$$

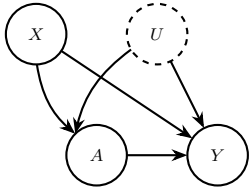


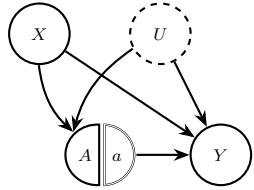


Overview

- However, for the remainder of the lectures, we will allow for unmeasured confounding between $A - Y$

$$Y(a) \not\perp\!\!\!\perp A \,|\, X$$





Overview

- Instrumental variable
- Difference-in-differences, Synthetic control
- Negative controls, Proximal causal inference
- ...

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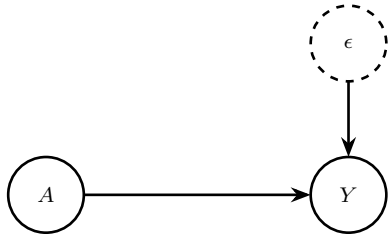
Linear Regression with Exogeneity

- Consider a linear model compatible with the graph

$$Y = \beta A + \epsilon \ , \quad \mathrm{Cov}(A, \epsilon) = 0 \ , \quad \mathrm{E}(A) = \mathrm{E}(\epsilon) = 0$$

- The OLS estimator is consistent:

$$\widehat{\beta} = \frac{\mathbb{P}(YA)}{\mathbb{P}(A^2)} \xrightarrow{P} \frac{\mathrm{E}(YA)}{\mathrm{E}(A^2)} = \beta + \frac{\mathrm{Cov}(A, \epsilon)}{\mathrm{Var}(A)} = \beta$$



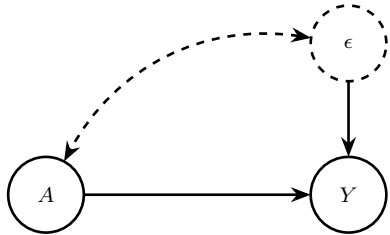
Linear Regression with Endogeneity

- Consider a linear model compatible with the graph

$$Y = \beta A + \epsilon \ , \quad \mathrm{Cov}(A, \epsilon) \neq 0 \ , \quad \mathrm{E}(A) = \mathrm{E}(\epsilon) = 0$$

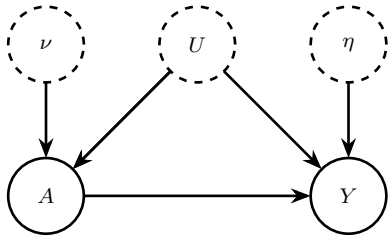
- The OLS estimator is inconsistent:

$$\widehat{\beta} = \frac{\mathbb{P}(YA)}{\mathbb{P}(A^2)} \xrightarrow{P} \frac{\mathrm{E}(YA)}{\mathrm{E}(A^2)} = \beta + \frac{\mathrm{Cov}(A, \epsilon)}{\mathrm{Var}(A)} \neq \beta$$



- Example: Omitted variable

$$Y = \beta A + \gamma U + \eta$$
$$A = \quad \delta U + \nu$$

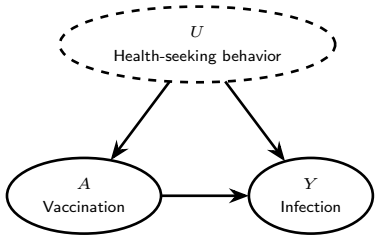


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Randomized Experiment vs. Randomized Encouragement Design

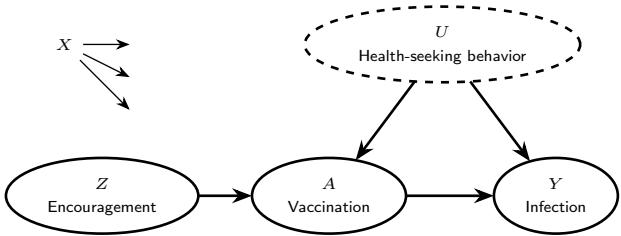






- Ideal scenario: Randomize flu vaccination directly
- Realistic scenario: Randomize encouragement to receive the flu vaccine, acknowledging that individuals may choose to comply or not

Randomized Experiment vs. Randomized Encouragement Design

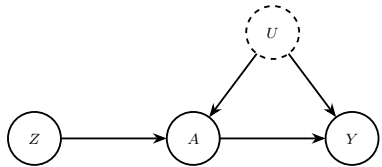




- Ideal scenario: Randomize flu vaccination directly
- Realistic scenario: Randomize encouragement to receive the flu vaccine, acknowledging that individuals may choose to comply or not

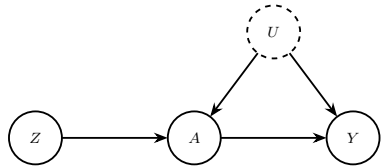
Example 2: Draft lottery for Vietnam war

- Hearst et al. (1986), Angrist (1990)
- $Z =$ draft lottery number (randomized)
- $A =$ military service
- $Y =$ mortality or earnings
- Men with low lottery numbers more likely to serve
- Lottery number unlikely to affect Y directly



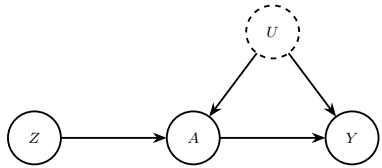
Example 3: Education on earnings

- Angrist and Krueger (1991)
- Z = quarter of birth (pseudo-random)
- A = years of schooling
- Y = earnings
- Policy rules tied school entry to age 6 and required school until age 16
- Quarter of birth influenced length of schooling
- Assumed to have no direct effect on earnings



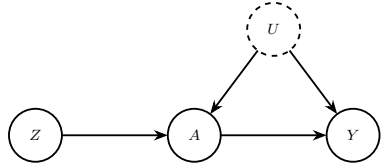
Example 4: Family Size and Employment

- Angrist and Evans (1998)
- $Z =$ sibling sex composition
- $A =$ family size
- $Y =$ mother's employment/work outcomes
- Parents with two same-sex children more likely to have a third
- Sibling sex composition is pseudo-random
- Assumed not to affect mother's work directly



Example 5: College Proximity and Wages

- Card (1993)
- $Z =$ growing up near a college (binary indicator)
- $A =$ years of schooling
- $Y =$ wages
- Potential issues:
  - Parents' location choice may not be random
  - Region may directly affect future wages
- Often cited, but possibly a weak or invalid IV

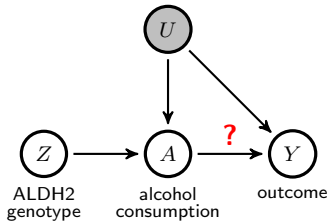


Example 6: Mendelian Randomization

- Chen et al. (2008)







Example 6: Mendelian Randomizationasdf

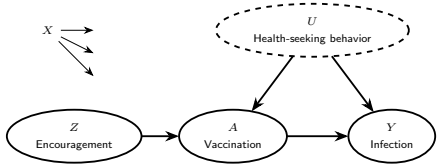
- Voight et al. (2012)
- Z = genetic variants (SNPs)
- A = HDL cholesterol levels
- Y = risk of heart attack
- Based on Mendel's laws: genes are randomly assigned
- Genes assumed to affect Y only via A
- Popular in biomedical research

Definition of an IV

- A variable Z qualifies as a valid IV if

Assumption (IV assumptions)

1. *(Exclusion Restriction)* $Y(z, a) = Y(a),\ a = 0, 1$
2. *(Exchangeability)* $Z \perp\!\!\!\perp \{Y(a), A(z)\}|X,\ a, z = 0, 1$
3. *(Relevance)* $Z \not\perp\!\!\!\perp A|X$



- In the randomized encouragement design:
  - The encouragement has no direct causal effect on infection, other than through vaccination
  - The encouragement is randomized and therefore independent of all pre-treatment variables
  - Receiving encouragement increases the likelihood of vaccination

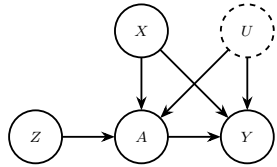
- Verifiability of the assumptions:
  - Relevance can be verified from the observed data; $\texttt{lm}(A \sim Z + X)$
  - ER and Exch cannot be verified from the observed data

- Exch sometimes written as $Z \perp\!\!\!\perp Y(z, a)|X$ for $a = 0, 1$; see Levis et al. (2024) for discussion of subtleties

Intent-to-Treat Approach

- ITT is defined by

$$\tau_{\text{ITT}} = \mathrm{E}\big[Y(A(z=1)) - Y(A(z=0))\big]$$



- One can consider Z as a treatment, and Y as an outcome

- The ITT analysis reduces to the previous ATT analysis, e.g.,

$$\widehat{\tau}_{\text{ITT}} = \mathrm{E}\big[\mathrm{E}(Y|Z=1,X) - \mathrm{E}(Y|Z=0,X)\big]$$

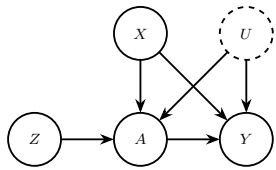
- However, the ITT does not identify the causal effect $A \to Y$

- While the definition of A remains fixed, the ITT can differ depending on how Z is defined
  - Example: A = vaccination; Z_1 = voucher worth \$10 for vaccination; Z_2 = Ferrari for vaccination.

Identification of $A \to Y$ Under the Three IV Assumptions

Assumption (IV assumptions)

1. *(Exclusion Restriction)* $Y(z,a) = Y(a),\ a = 0,1$
2. *(Exchangeability)* $Z \perp\!\!\!\perp \{Y(a), A(z)\}|X,\ a,z = 0,1$
3. *(Relevance)* $Z \not\perp\!\!\!\perp A|X$



- Assume binary Z, A, Y
- From the observed data, we can learn the following 7 parameters

$$\Pr(Z=1)\ , \quad \Pr(A=1|Z=z)\ , \quad \Pr(Y=1|Z=z,A=a)\ , \quad z,a=0,1$$

- Note that $Z \to A \leftarrow U \to Y$ implies $Y \not\perp\!\!\!\perp Z \,|\, A$

Identification of $A \to Y$ Under the Three IV Assumptions

- The full data $\{Y(a=1) - Y(a=0), A(z=1), A(z=0), Z\}$ for $a = 0, 1$ can be expressed as

$$\begin{aligned}
&\Pr\{Y(a=1) - Y(a=0) = y, A(1) = a_1, A(0) = a_0, Z = z\} \\
&= \underbrace{\Pr\{Y(a=1) - Y(a=0) = y, A(1) = a_1, A(0) = a_0\}}_{=\text{sum to 1}} \Pr(Z = z)
\end{aligned}$$

- We have 12 parameters characterizing $\{Y(a=1) - Y(a=0), A(z=1), A(z=0), Z\}$:

$$\Pr(Z=1)\ , \quad \Pr\{Y(a=1) - Y(a=0) = y, A(1) = a_1, A(0) = a_0\}\ , \quad y \in -1, 0, 1,\ a_1 = 0, 1,\ a_0 = 0, 1$$

- # free parameters = 12 > 7 = # identified parameters

- Need the **fourth IV assumption** in order to identify the causal effect

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Latent Compliance Types

Define compliance behavior (U) based on $(A(1), A(0))$:

- Always takers (a): $A(1) = 1, A(0) = 1$
- Compliers (c): $A(1) = 1, A(0) = 0$
- Defiers (d): $A(1) = 0, A(0) = 1$
- Never takers (n): $A(1) = 0, A(0) = 0$

		$M(1)$	
		0	1
$M(0)$	0	NT	CO
	1	DF	AT

Using the law of total probability:

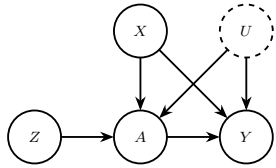
$$\tau_Y = \left[\begin{array}{l} \mathrm{E}[Y(1) - Y(0) \mid U = \mathtt{a}] \Pr(U = \mathtt{a}) \\ \quad + \mathrm{E}[Y(1) - Y(0) \mid U = \mathtt{c}] \Pr(U = \mathtt{c}) \\ \quad + \mathrm{E}[Y(1) - Y(0) \mid U = \mathtt{d}] \Pr(U = \mathtt{d}) \\ \quad + \mathrm{E}[Y(1) - Y(0) \mid U = \mathtt{n}] \Pr(U = \mathtt{n}) \end{array} \right]$$

Monotonicity Assumption

- Angrist et al. (1996):

Assumption (Monotonicity)

$\Pr(U = \mathbf{d}) = 0$ *or* $A(1) \geqslant A(0)$ *for all units*



- This rules out defiers
- There is no way to encode this assumption in the causal graph
- The monotonicity assumption coupled with exclusion restriction $Y(1, a) = Y(0, a)$ implies that

$$Y(z = 1) - Y(z = 0) = Y(1, A(1)) - Y(0, A(0)) = \begin{cases} 0 & \text{if } U_i = \mathtt{a} \\ 0 & \text{if } U_i = \mathtt{n} \\ Y(d = 1) - Y(d = 0) & \text{if } U_i = \mathtt{c} \end{cases}$$

Definition of the LATE

$$\tau_Y = \begin{bmatrix} \mathrm{E}[Y(1) - Y(0) \mid U = \mathtt{a}] \Pr(U = \mathtt{a}) \\ +\mathrm{E}[Y(1) - Y(0) \mid U = \mathtt{c}] \Pr(U = \mathtt{c}) \\ +\mathrm{E}[Y(1) - Y(0) \mid U = \mathtt{d}] \Pr(U = \mathtt{d}) \\ +\mathrm{E}[Y(1) - Y(0) \mid U = \mathtt{n}] \Pr(U = \mathtt{n}) \end{bmatrix}, \quad \tau_A = \begin{bmatrix} \mathrm{E}[A(1) - A(0) \mid U = \mathtt{a}] \Pr(U = \mathtt{a}) \\ +\mathrm{E}[A(1) - A(0) \mid U = \mathtt{c}] \Pr(U = \mathtt{c}) \\ +\mathrm{E}[A(1) - A(0) \mid U = \mathtt{d}] \Pr(U = \mathtt{d}) \\ +\mathrm{E}[A(1) - A(0) \mid U = \mathtt{n}] \Pr(U = \mathtt{n}) \end{bmatrix}$$

Under monotonicity and exclusion restriction:

$$\tau_Y = \mathrm{E}[Y(a = 1) - Y(a = 0) \mid U = \mathtt{c}] \Pr(U = \mathtt{c})$$
$$\tau_A = \Pr(U = \mathtt{c})$$

Therefore, the Complier Average Causal Effect (CACE) or Local Average Treatment Effect (LATE) is

$$\tau_{\mathsf{LATE}} = \mathrm{E}[Y(a = 1) - Y(a = 0) \mid U = \mathtt{c}] = \frac{\tau_Y}{\tau_A}$$

Identification of the LATE

Assumption (IV assumptions + Monotonicity)

1. *(Exclusion Restriction)* $Y(z,a) = Y(a),\ a = 0,1$
2. *(Exchangeability)* $Z \perp\!\!\!\perp \{Y(a), A(z)\}|X,\ a,z = 0,1$
3. *(Relevance)* $Z \not\perp\!\!\!\perp A|X$
4. *(Monotonicity)* $A(1) \geqslant A(0)$

- Under the IV assumptions and monotonicity, the LATE is identified by

$$\tau_{\mathsf{LATE}} = \mathrm{E}[Y(a=1) - Y(a=0) \mid U = \mathsf{c}] = \frac{\tau_Y}{\tau_A} = \frac{\mathrm{E}_X\big[\mathrm{E}(Y \,\big|\, Z=1,X) - \mathrm{E}(Y \,\big|\, Z=0,X)\big]}{\mathrm{E}_X\big[\mathrm{E}(A \,\big|\, Z=1,X) - \mathrm{E}(A \,\big|\, Z=0,X)\big]}$$

- Joshua Angrist and Guido Imbens were awarded the 2021 Nobel Prize in Economic Sciences for their groundbreaking methodological contributions to the analysis of causal relationships, particularly through the use of IV in observational studies and "natural experiments"

Assumption (IV assumptions + Monotonicity)

1. *(Exclusion Restriction)* $Y(z,a) = Y(a)$, $a = 0, 1$
2. *(Exchangeability)* $Z \perp\!\!\!\perp \{Y(a), A(z)\}|X$, $a, z = 0, 1$
3. *(Relevance)* $Z \not\perp\!\!\!\perp A|X$
4. *(Monotonicity)* $A(1) \geqslant A(0)$

$$
\begin{aligned}
\tau_Y &= \mathrm{E}[Y(a=1) - Y(a=0) \mid U = \mathsf{c}] \Pr(U = \mathsf{c}) \\
&= \mathrm{E}\big[\{Y(a=1) - Y(a=0)\}\mathbb{1}\{A(z=1) = 1, A(z=0) = 0\}\big] \\
&= \mathrm{E}\big[\{Y(z=1) - Y(z=0)\}\{A(z=1) - A(z=0)\}\big] \\
&= \mathrm{E}\big[Y(z=1, A(z=1)=1) - Y(z=0, A(z=1)=0)\big] \\
&= \mathrm{E}_X\left[\begin{array}{l} \mathrm{E}\big[Y(z=1, A(z=1)=1) \\ \quad -Y(z=0, A(z=0)=0) \,\big|\, X\big] \end{array}\right] \\
&= \mathrm{E}_X\left[\begin{array}{l} \mathrm{E}\big[Y(z=1, A(z=1)=1) \,\big|\, X\big] \\ \quad -\mathrm{E}\big[Y(z=0, A(z=0)=0) \,\big|\, X\big] \end{array}\right] \\
&= \mathrm{E}_X\left[\begin{array}{l} \mathrm{E}\big[Y(z=1, A(z=1)=1) \,\big|\, Z=1, X\big] \\ \quad -\mathrm{E}\big[Y(z=0, A(z=0)=0) \,\big|\, Z=0, X\big] \end{array}\right] \\
&= \mathrm{E}_X\big[\mathrm{E}(Y \,|\, Z=1, X) - \mathrm{E}(Y \,|\, Z=0, X)\big]
\end{aligned}
$$

$$
\begin{aligned}
\tau_A &= \Pr(U = \mathsf{c}) \\
&= \mathrm{E}\big[A(z=1) - A(z=0)\big] \\
&= \mathrm{E}_X\Big[\mathrm{E}\big[A(z=1) - A(z=0) \,\big|\, X\big]\Big] \\
&= \mathrm{E}_X\left[\begin{array}{l} \mathrm{E}\big[A(z=1) \,\big|\, Z=1, X\big] \\ \quad -\mathrm{E}\big[A(z=0) \,\big|\, Z=0, X\big] \end{array}\right] \\
&= \mathrm{E}_X\big[\mathrm{E}(A \,|\, Z=1, X) - \mathrm{E}(A \,|\, Z=0, X)\big]
\end{aligned}
$$

Estimation of the LATE

- The LATE is identified by

$$\tau_{\mathsf{LATE}} = \mathrm{E}[Y(a=1) - Y(a=0) \mid U = \mathsf{c}] = \frac{\tau_Y}{\tau_A} = \frac{\mathrm{E}_X\big[\mathrm{E}(Y \,|\, Z=1, X) - \mathrm{E}(Y \,|\, Z=0, X)\big]}{\mathrm{E}_X\big[\mathrm{E}(A \,|\, Z=1, X) - \mathrm{E}(A \,|\, Z=0, X)\big]}$$

- We estimate the LATE by a simple ratio:

$$\widehat{\tau}_{\mathsf{LATE}} = \frac{\widehat{\tau}_Y}{\widehat{\tau}_A}$$

where $\widehat{\tau}_Y$ and $\widehat{\tau}_A$ can be obtained from the IPW/OR/AIPW estimators, e.g.,

$$\widehat{\tau}_Y = \frac{1}{N}\sum_{i=1}^{N}\big[\widehat{\mathrm{E}}(Y|Z=1, X_i) - \widehat{\mathrm{E}}(Y|Z=0, X_i)\big]$$

- This is called the Wald estimator or IV estimator
- One can use bootstrap for obtain the SE

EIF of the LATE

$$\tau_{\mathsf{LATE}} = \frac{\tau_Y}{\tau_A} = \frac{\mathrm{E}_X\big[\mathrm{E}(Y \,|\, Z=1,X) - \mathrm{E}(Y \,|\, Z=0,X)\big]}{\mathrm{E}_X\big[\mathrm{E}(A \,|\, Z=1,X) - \mathrm{E}(A \,|\, Z=0,X)\big]}$$

- The gradient of the LATE in the parametric submodel is

$$\left.\frac{\partial \tau_{\mathsf{LATE}}(\eta)}{\partial \eta}\right|_{\eta=\eta^*} = \frac{\nabla\tau_Y(\eta^*) - \tau_{\mathsf{LATE}} \cdot \nabla\tau_A(\eta^*)}{\tau_A} = \mathrm{E}\left[\frac{\mathsf{IF}_Y(O;\tau_Y) - \tau_{\mathsf{LATE}} \cdot \mathsf{IF}_A(O;\tau_A)}{\tau_A} \times S(O)\right]$$

  where

$$\mathsf{IF}_Y(O) = \frac{Z\{Y - \mathrm{E}(Y|Z=1,X)\}}{\Pr(Z=1|X)} - \frac{(1-Z)\{Y - \mathrm{E}(Y|Z=0,X)\}}{\Pr(Z=0|X)} + \mathrm{E}(Y|Z=1,X) - \mathrm{E}(Y|Z=0,X)$$

$$\mathsf{IF}_A(O) = \frac{Z\{A - \mathrm{E}(A|Z=1,X)\}}{\Pr(Z=1|X)} - \frac{(1-Z)\{A - \mathrm{E}(A|Z=0,X)\}}{\Pr(Z=0|X)} + \mathrm{E}(A|Z=1,X) - \mathrm{E}(A|Z=0,X)$$

- Therefore, an IF for the LATE is

$$\mathsf{IF}_{\mathsf{LATE}}(O) = \frac{\mathsf{IF}_Y(O) - \tau_{\mathsf{LATE}} \cdot \mathsf{IF}_A(O)}{\tau_A}$$

- This IF is the EIF for the LATE in the nonparametric model

Falsifiability of the IV and Monotonicity Assumptions

- Suppose that the 3 IV and the monotonicity assumptions are satisfied. Then, Kitagawa (2015) showed:

$$P(Y = y, A = a|Z = 1, X) \geqslant P(Y = y, A = a|Z = 0, X) \ , \quad \forall y, \ \forall a = 0, 1 \tag{1}$$

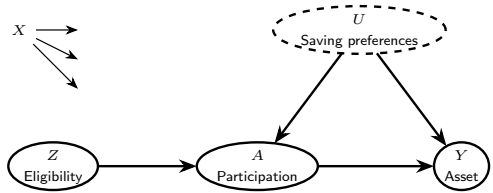
- If (1) does not hold, then at least one of the assumptions is violated, though we cannot determine which one

Example: 401(k)

- `DoubleML::fetch_401k`

- 401(k) is an employer-sponsored retirement savings plan that allows employees to contribute a portion of their wages to individual investment accounts

Role	Variable	Description
X	age	age
	db	= 1 if individual has defined benefit pension
	educ	years of education
	fsize	family size
	hown	= 1 if home owner
	inc	income
	marr	= 1 if married
	pira	= 1 if individual participates in IRA plan
	twoearn	= 1 if two-earner household
Z	e401	= 1 if employer offers 401(k)
A	p401	= 1 if individual participates in a 401(k) plan
Y	net_tfa	net total financial assets



- Effect of 401(k) participation on accumulated assets using eligibility as an IV

Example: 401(k)

- **Exclusion**
- 401(k) eligibility affects net financial assets only through participation and has no direct effect

- **Exchangeability**
- 401(k) eligibility is determined by observed, exogenous factors (e.g., income) and not by unobserved, endogenous factors (e.g., saving preferences)

Example: 401(k)

- **Relevance**
- Subject-matter knowledge:
  - One can only participate in a 401(k) $(A = 1)$ if eligible $(Z = 1)$

$$\Pr(A = 1|Z = 1, X) > \Pr(A = 1|Z = 0, X) = 0$$

- Empirical check:

```
library(DoubleML)
data <- fetch_401k(return_type = "data.table", instrument = TRUE)

table( data$e401, data$p401 )

       0    1
  0 6233    0
  1 1088 2594
```

- Note: Marginal association does not imply the conditional association

Example: 401(k)

- **Monotonicity**
- One can only participate in a 401(k) $(A = 1)$ if eligible $(Z = 1)$ i.e., $A(Z = 0) = 0$
- This means that DF=$\{A(Z = 1) = 0, A(Z = 0) = 1\} = \varnothing$
- In fact, this also means that AF=$\{A(Z = 1) = 1, A(Z = 0) = 1\} = \varnothing$
- **One-sided noncompliance**: either CO or NT

Example: 401(k)

```
LATE.AIPW <- function(data){
  PS <- glm(e401~(age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data,family="binomial")
  mu.A <- glm(p401~(e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data,family="binomial")
  mu.Y <- lm(net_tfa~(e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data)

  data1 <- data0 <- data; data1$e401 <- 1; data0$e401 <- 0

  PS.Est  <- predict(PS,type="response")
  mu.A.Z1.Est <- predict(mu.A,data1,type="response"); mu.A.Z0.Est <- predict(mu.A,data0,type="response")
  mu.Y.Z1.Est <- predict(mu.Y,data1); mu.Y.Z0.Est <- predict(mu.Y,data0)

  Z <- data$e401; A <- data$p401; Y <- data$net_tfa

  numerator <- mean( Z*(Y-mu.Y.Z1.Est)/PS.Est-(1-Z)*(Y-mu.Y.Z0.Est)/(1-PS.Est)+(mu.Y.Z1.Est-mu.Y.Z0.Est) )
  denominator <- mean( Z*(A-mu.A.Z1.Est)/PS.Est-(1-Z)*(A-mu.A.Z0.Est)/(1-PS.Est)+(mu.A.Z1.Est-mu.A.Z0.Est) )
  return( numerator/denominator )
}

LATE.AIPW.Estimate <- LATE.AIPW(data)
N <- nrow(data);  LATE.AIPW.Boot <- sapply(1:100,function(ii){LATE.AIPW(data[sample(N,N,replace=TRUE),])})
c(LATE.AIPW.Estimate, sd(LATE.AIPW.Boot), quantile(LATE.AIPW.Boot,c(0.025,0.975)))
```

- $\widehat{\tau}_{\mathsf{LATE}} = 8209.923$, $\mathsf{SE} = 2515.916$, $95\%\ \mathsf{CI} = [2588.545, 12207.809]$

Outline

1. Overview
2. Review: Linear Regression
3. Instrumental Variable
4. LATE/CACE
5. ATT
6. ATE
7. TSLS
8. Nonparametric IV
9. Additional Topics
10. Assignment

No Current-Treatment Value Interaction

- Robins (1994): no current-treatment value interaction (NCTI)

Assumption (No Current-Treatment Value Interaction)

$$\mathrm{E}\{Y(a=1) - Y(a=0) \,|\, A=1, Z=1, X\} = \mathrm{E}\{Y(a=1) - Y(a=0) \,|\, A=1, Z=0, X\}$$

- The causal effect $A \to Y$ is identical among subjects who actually received A regardless of assigned treatment Z
- Does not require the monotonicity assumption
  - the relation between Z and A can be increasing for some individuals and decreasing for others
- But the assumption requires ruling out the possibility of effect heterogeneity w.r.t. Z
- The following model is implied by the following structural model:

$$\mathrm{E}\{Y(a) \,|\, A=1, U, Z\} = \beta_0 + \beta_a a + \beta_u(U)$$

Identification of the ATT

Assumption (IV assumptions + NCTI)

1. *(Exclusion Restriction)* $Y(z,a) = Y(a),\ a = 0,1$
2. *(Exchangeability)* $Z \perp\!\!\!\perp \{Y(a), A(z)\}|X,\ a, z = 0,1$
3. *(Relevance)* $Z \not\perp\!\!\!\perp A|X$
4. *(NCTI)* $\mathrm{E}\{Y(a=1) - Y(a=0) \,|\, A=1, Z=1, X\} = \mathrm{E}\{Y(a=1) - Y(a=0) \,|\, A=1, Z=0, X\}$

- Under the IV assumptions and NCTI, we have

$$\tau_{\mathsf{ATT}} = \mathrm{E}\big\{Y(a=1) - Y(a=0) \,\big|\, A=1\big\} = \mathrm{E}_X\left[\frac{\mathrm{E}(Y|Z=1,X) - \mathrm{E}(Y|Z=0,X)}{\mathrm{E}(A|Z=1,X) - \mathrm{E}(A|Z=0,X)} \,\middle|\, A=1\right]$$

Assumption (IV assumptions + NCTI)

1. *(Exclusion Restriction)* $Y(z,a) = Y(a),\ a = 0,1$
2. *(Exchangeability)* $Z \perp\!\!\!\perp \{Y(a), A(z)\}|X,\ a, z = 0,1$
3. *(Relevance)* $Z \not\perp\!\!\!\perp A|X$
4. *(NCTI)* $\mathrm{E}\{Y(a=1) - Y(a=0) \,|\, A=1, Z=1, X\} = \mathrm{E}\{Y(a=1) - Y(a=0) \,|\, A=1, Z=0, X\}$

$$\tau_{\mathsf{ATT}}(X) := \mathrm{E}\big\{Y(a=1) - Y(a=0) \,\big|\, A=1, X\big\} \overset{\star}{=} \mathrm{E}\big\{Y(a=1) - Y(a=0) \,\big|\, A=1, Z, X\big\} \tag{2}$$

where $\star$ is from the NCTI assumption. We also have

$$\mathrm{E}\big\{Y - \tau_{\mathsf{ATT}}(X)A \,\big|\, A, Z, X\big\} = \begin{cases} \mathrm{E}\big\{Y(a=0) \,\big|\, A=0, Z, X\big\} & \text{if } A=0 \\ \mathrm{E}\big\{Y(a=1) - \tau_{\mathsf{ATT}}(X) \,\big|\, A=1, Z, X\big\} & \text{if } A=1 \end{cases} \overset{(2)}{=} \mathrm{E}\big\{Y(a=0) \,\big|\, A, Z, X\big\}$$

Integrating out A, we have

$$\mathrm{E}\big\{Y - \tau_{\mathsf{ATT}}(X)A \,\big|\, Z, X\big\} = \mathrm{E}\big\{Y(a=0) \,\big|\, Z, X\big\} \overset{\mathsf{Exch}}{=} \mathrm{E}\big\{Y(a=0) \,\big|\, X\big\} \ \Rightarrow\ \mathrm{E}\big\{Y - \tau_{\mathsf{ATT}}(X)A \,\big|\, Z, X\big\} = \mathrm{E}\big\{Y - \tau_{\mathsf{ATT}}(X)A \,\big|\, X\big\}$$

This means that $\mathrm{Cov}\big\{Z, Y - \tau_{\mathsf{ATT}}(X)A \,\big|\, X\big\} = 0$, i.e.,

$$\mathrm{Cov}(Z, Y|X) = \tau_{\mathsf{ATT}}(X)\mathrm{Cov}(Z, A|X) \quad \Rightarrow \quad \tau_{\mathsf{ATT}}(X) = \frac{\mathrm{Cov}(Z, Y|X)}{\mathrm{Cov}(Z, A|X)} = \frac{\mathrm{E}(Y|Z=1, X) - \mathrm{E}(Y|Z=0, X)}{\mathrm{E}(A|Z=1, X) - \mathrm{E}(A|Z=0, X)}$$

Marginalizing out X over $A = 1$, we have

$$\tau_{\mathsf{ATT}} = \mathrm{E}_X\{\tau_{\mathsf{ATT}}(X)|A=1\} = \mathrm{E}_X\left[\frac{\mathrm{E}(Y|Z=1, X) - \mathrm{E}(Y|Z=0, X)}{\mathrm{E}(A|Z=1, X) - \mathrm{E}(A|Z=0, X)} \,\middle|\, A=1\right]$$

Estimation of the ATT

- The ATT is identified by

$$\tau_{\text{ATT}} = \mathrm{E}_X \left[\frac{\mathrm{E}(Y|Z=1,X) - \mathrm{E}(Y|Z=0,X)}{\mathrm{E}(A|Z=1,X) - \mathrm{E}(A|Z=0,X)} \,\middle|\, A=1 \right]$$

- We estimate the ATT by a simple ratio:

$$\widehat{\tau}_{\text{ATT}} = \frac{1}{\sum_{i=1}^N A_i} \sum_{i=1}^N A_i \left[\frac{\widehat{\mathrm{E}}(Y|Z=1,X_i) - \widehat{\mathrm{E}}(Y|Z=0,X_i)}{\widehat{\mathrm{E}}(A|Z=1,X_i) - \widehat{\mathrm{E}}(A|Z=0,X_i)} \right]$$

- One can use bootstrap for obtain the SE

Multiplicative IV model

- Assume a weaker exchangeability: $Z \perp\!\!\!\perp U|X$ where $(A, Z) \perp\!\!\!\perp Y(a) \,|\, (U, X)$, $a = 0, 1$
- Hernán and Robins (2006): Multiplicative IV model

Assumption (Multiplicative IV Model)

$$\Pr(A = 1|Z, U, X) = \alpha_z(Z, X)\alpha_u(U, X)$$

- There is no multiplicative interaction between Z and U in causing A
- In fact, the MIV implies NCTI

$$
\begin{aligned}
f(U|A,Z,X) &= \frac{f(A|U,Z,X)f(U|Z,X)}{\int f(A|u,Z,X)f(u|Z,X)\,du} \\
&= \frac{f(A|U,Z,X)f(U,X)}{\int f(A|u,Z,X)f(u|X)\,du} \\
&= \frac{\frac{f(A|U,Z,X)}{f(A|U,Z=0,X)}f(A|U,Z=0,X)f(U|X)}{\int \frac{f(A|u,Z,X)}{f(A|u,Z=0,X)}f(A|u,Z=0,X)f(u|X)\,du} \\
&= \frac{f(A|U,Z=0,X)f(U|X)}{\int f(A|u,Z=0,X)f(u|X)\,du} \\
&= g(U,X)
\end{aligned}
$$

Thus,

$$
\begin{aligned}
\mathrm{E}\{Y(a=1)-Y(a=0)|A=1,Z,X\} &= \mathrm{E}\big[\mathrm{E}\{Y(a=1)-Y(a=0)|A=1,Z,U,X\}\,\big|\,A=1,Z,X\big] \\
&= \mathrm{E}\big[\mathrm{E}\{Y(a=1)-Y(a=0)|U,X\}\,\big|\,A=1,Z,X\big] \\
&= \int \mathrm{E}\{Y(a=1)-Y(a=0)|U=u,X\}f(U=u|A,Z,X)\,du \\
&= \int \mathrm{E}\{Y(a=1)-Y(a=0)|U=u,X\}g(u,X)\,du \\
&= h(X)
\end{aligned}
$$

Identification of the ATT under the MIV

Assumption (IV assumptions + MIV)

1. *(Exclusion Restriction)* $Y(z,a) = Y(a)$, $a = 0, 1$
2. *(Exchangeability)* $Z \perp\!\!\!\perp U|X$ *where* $(A, Z) \perp\!\!\!\perp Y(a) \,|\, (U, X)$, $a = 0, 1$
3. *(Relevance)* $Z \not\perp\!\!\!\perp A|X$
4. *(MIV)* $\Pr(A = 1|Z, U, X) = \alpha_z(Z, X)\alpha_u(U, X)$

- Under the IV assumptions and MIV, we have

$$\tau_{\text{ATT}} = \mathrm{E}\big\{Y(a=1) - Y(a=0) \,\big|\, A = 1\big\} = \mathrm{E}_X\left[\frac{\mathrm{E}(Y|Z=1,X) - \mathrm{E}(Y|Z=0,X)}{\mathrm{E}(A|Z=1,X) - \mathrm{E}(A|Z=0,X)} \,\middle|\, A = 1\right]$$

Identification of the ATT under the MIV

- In fact, we only need the exchangeability for $Y(0)$:

Assumption (IV assumptions + MIV)

1. (Exclusion Restriction)	$Y(z,a) = Y(a),\ a = 0, 1$
2. (Weak Exchangeability)	$Z \perp\!\!\!\perp U \mid X$ *where* $(A, Z) \perp\!\!\!\perp Y(0) \,\big|\, (U, X)$
3. (Relevance)	$Z \not\perp\!\!\!\perp A \mid X$
4. (MIV)	$\Pr(A = 1 \mid Z, U, X) = \alpha_z(Z, X)\alpha_u(U, X)$

- Liu et al. (2025): Under the IV assumptions and MIV, we have

$$\begin{aligned}\tau_{\text{ATT}} &= \mathrm{E}\big\{Y(a=1) - Y(a=0) \,\big|\, A = 1\big\} \\ &= \mathrm{E}(Y|A=1) - \mathrm{E}_X\left[- \frac{\mathrm{E}\{Y(1-A)|Z=1, X\} - \mathrm{E}\{Y(1-A)|Z=0, X\}}{\mathrm{E}(A|Z=1, X) - \mathrm{E}(A|Z=0, X)} \Big| A = 1 \right]\end{aligned}$$

- Only need a Wald ratio statistic for $\mathrm{E}\big\{Y(a=0) \,\big|\, A = 1\big\}$; **single-arm Wald ratio statistic**

- See Liu et al. (2025) for the EIF of the ATT under the MIV

Example: 401(k)

```
ATT.HR <- function(data){
  mu.A <- glm(p401~(e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data,family="binomial")
  mu.Y <- lm(net_tfa~(e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data)

  data1 <- data0 <- data; data1$e401 <- 1; data0$e401 <- 0

  mu.A.Z1.Est <- predict(mu.A,data1,type="response"); mu.A.Z0.Est <- predict(mu.A,data0,type="response")
  mu.Y.Z1.Est <- predict(mu.Y,data1); mu.Y.Z0.Est <- predict(mu.Y,data0)

  Z <- data$e401; A <- data$p401; Y <- data$net_tfa

  numerator <- (mu.Y.Z1.Est-mu.Y.Z0.Est)
  denominator <- (mu.A.Z1.Est-mu.A.Z0.Est)
  return( mean((numerator/denominator)[data$p401==1]) )
}
```

Example: 401(k)

```
ATT.Liu <- function(data){
  mu.A <- glm(p401~(e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data,family="binomial")
  mu.Y.given.A0 <- lm(net_tfa~(e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data[data$p401==0,])

  data1 <- data0 <- data; data1$e401 <- 1; data0$e401 <- 0

  mu.A.Z1.Est <- predict(mu.A,data1,type="response"); mu.A.Z0.Est <- predict(mu.A,data0,type="response")
  mu.Y.given.A0.Z1.Est <- predict(mu.Y.given.A0,data1); mu.Y.given.A0.Z0.Est <- predict(mu.Y.given.A0,data0)

  Z <- data$e401; A <- data$p401; Y <- data$net_tfa

  numerator <- -(mu.Y.given.A0.Z1.Est*(1-mu.A.Z1.Est)-mu.Y.given.A0.Z0.Est*(1-mu.A.Z0.Est))
  denominator <- (mu.A.Z1.Est-mu.A.Z0.Est)
  return( mean(data$net_tfa[data$p401==1]) - mean((numerator/denominator)[data$p401==1]) )
}
```

Example: 401(k)

```
ATT.HR.Estimate <- ATT.HR(data)
N <- nrow(data)
ATT.HR.Boot <- sapply(1:100,function(ii){ATT.HR(data[sample(N,N,replace=TRUE),])})
c(ATT.HR.Estimate, sd(ATT.HR.Boot), quantile(ATT.HR.Boot,c(0.025,0.975)))
```

```
ATT.Liu.Estimate <- ATT.Liu(data)
N <- nrow(data)
ATT.Liu.Boot <- sapply(1:100,function(ii){ATT.Liu(data[sample(N,N,replace=TRUE),])})
c(ATT.Liu.Estimate, sd(ATT.Liu.Boot), quantile(ATT.Liu.Boot,c(0.025,0.975)))
```

- MIV with strong exchangeability: $\widehat{\tau}_{\text{ATT}} = 13628.697$, SE $= 2941.887$, 95% CI $= (8724.051, 19406.361)$
- MIV with weak exchangeability: $\widehat{\tau}_{\text{ATT}} = 14368.641$, SE $= 2624.231$, 95% CI $= (9708.681, 19750.706)$
- Pretty similar to each other

Outline

1. Overview
2. Review: Linear Regression
3. Instrumental Variable
4. LATE/CACE
5. ATT
6. ATE
7. TSLS
8. Nonparametric IV
9. Additional Topics
10. Assignment

No Additive Modification

- Assume a weaker exchangeability: $Z \perp\!\!\!\perp U|X$ where $(A, Z) \perp\!\!\!\perp Y(a) \,|\, (U, X)$, $a = 0, 1$

- Wang and Tchetgen Tchetgen (2018):

Assumption (No Additive Modification)

Suppose that either (a) or (b) holds

(a) $\rho_y(U, X) = \mathrm{E}\{Y(1) - Y(0)|U, X\}$ *does not depend on* U*;*

(b) $\rho_a(U, X) = \mathrm{E}(A|Z = 1, U, X) - \mathrm{E}(A|Z = 0, U, X)$ *does not depend on* U

- (a) $=$ no latent heterogeneity of the ATE

- (b) $=$ no unmeasured confounder of $A \to Y$

Identification of the ATE under the No Additive Modification

Assumption (IV assumptions + No Additive Modification)

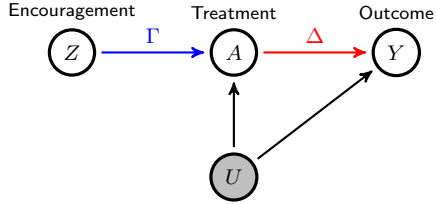
1. (Exclusion Restriction)	$Y(z,a) = Y(a)$, $a = 0, 1$
2. (Exchangeability)	$Z \perp\!\!\!\perp \{Y(a), A(z)\}|X$, $a, z = 0, 1$
3. (Relevance)	$Z \not\perp\!\!\!\perp A|X$
4. (No Add. Mod.)	*4-1:* $\rho_y(X) = \mathrm{E}\{Y(1) - Y(0)|U, X\}$ *or* *4-2:* $\rho_a(X) = \mathrm{E}(A|Z = 1, U, X) - \mathrm{E}(A|Z = 0, U, X)$

- Wang and Tchetgen Tchetgen (2018): under these assumptions, we have

$$\tau_{\mathsf{ATE}} = \mathrm{E}\big\{Y(a = 1) - Y(a = 0)\big\} = \mathrm{E}_X\left[\frac{\mathrm{E}(Y|Z = 1, X) - \mathrm{E}(Y|Z = 0, X)}{\mathrm{E}(A|Z = 1, X) - \mathrm{E}(A|Z = 0, X)}\right]$$

- The result under 4-1 is not surprising; the average causal effect is the same for each value of U
- The result under 4-2 surprising as it does not place any restriction on the effect heterogeneity of $A \to Y$
  - it only restricts the first stage relation $Z \to A$
- See Wang and Tchetgen Tchetgen (2018) for the EIF of the ATE

Intuitive Proof



- Goal is to learn $\mathrm{E}(\Delta_i)$; note that

$$\mathsf{ITE}_i(Z \to Y) = \Gamma_i \times \Delta_i$$

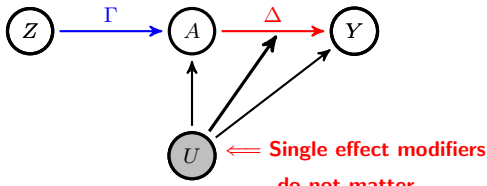
$$\mathsf{ATE}(Z \to Y) = \mathrm{E}(\Gamma_i \times \Delta_i) \qquad \text{Taking Expectation}$$

- We can identify $\mathsf{ATE}(Z \to Y)$ and $\mathrm{E}(\Gamma_i)$

- If $\Gamma_i \perp\!\!\!\perp \Delta_i$, we have

$$\mathsf{ATE}(Z \to Y) = \mathrm{E}(\Gamma_i \times \Delta_i) = \mathrm{E}(\Gamma_i)\mathrm{E}(\Delta_i) \quad \Rightarrow \quad \mathrm{E}(\Delta_i) = \frac{\mathsf{ATE}(Z \to Y)}{\mathrm{E}(\Gamma_i)}$$

Intuitive Proof

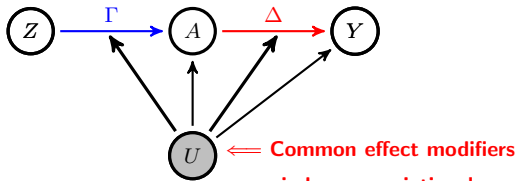


- $\Gamma_i \perp\!\!\!\perp \Delta_i$ if

$$\Gamma_i = \mathrm{E}(A|Z=1,U,X) - \mathrm{E}(A|Z=0,U,X) \quad \text{does not depend on } U; \quad \text{or}$$

$$\Delta_i = \mathrm{E}\{Y(1) - Y(0)|U,X\} \quad \text{does not depend on } U$$

Intuitive Proof



- $\Gamma_i \perp\!\!\!\perp \Delta_i$ if

$\Gamma_i = \mathrm{E}(A|Z=1,U,X) - \mathrm{E}(A|Z=0,U,X)$ does not depend on U; or

$\Delta_i = \mathrm{E}\{Y(1) - Y(0)|U,X\}$ does not depend on U

Comparison between the LATE, ATT, ATE

- Under their respective IV assumptions, the LATE, ATT, ATE are identified by:

$$\tau_{\text{LATE}} = \frac{\mathrm{E}_X\big[\mathrm{E}(Y \,|\, Z=1,X) - \mathrm{E}(Y \,|\, Z=0,X)\big]}{\mathrm{E}_X\big[\mathrm{E}(A \,|\, Z=1,X) - \mathrm{E}(A \,|\, Z=0,X)\big]}$$

$$\tau_{\text{ATT}} = \mathrm{E}_X\left[\frac{\mathrm{E}(Y|Z=1,X) - \mathrm{E}(Y|Z=0,X)}{\mathrm{E}(A|Z=1,X) - \mathrm{E}(A|Z=0,X)} \,\middle|\, A=1\right]$$

$$\tau_{\text{ATE}} = \mathrm{E}_X\left[\frac{\mathrm{E}(Y|Z=1,X) - \mathrm{E}(Y|Z=0,X)}{\mathrm{E}(A|Z=1,X) - \mathrm{E}(A|Z=0,X)}\right]$$

- Note that all effects are expressed using the Wald ratio statistic, but
  - LATE: ratio of expectations
  - ATT/ATE: expectation of a ratio

Example: 401(k)

```
ATE.WTT <- function(data){
  mu.A <- glm(p401~(e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data,family="binomial")
  mu.Y <- lm(net_tfa~(e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2, data=data)

  data1 <- data0 <- data; data1$e401 <- 1; data0$e401 <- 0

  mu.A.Z1.Est <- predict(mu.A,data1,type="response"); mu.A.Z0.Est <- predict(mu.A,data0,type="response")
  mu.Y.Z1.Est <- predict(mu.Y,data1); mu.Y.Z0.Est <- predict(mu.Y,data0)

  Z <- data$e401; A <- data$p401; Y <- data$net_tfa

  numerator <- (mu.Y.Z1.Est-mu.Y.Z0.Est)
  denominator <- (mu.A.Z1.Est-mu.A.Z0.Est)
  return( mean(numerator/denominator) )
}
```

- This is similar to the estimator under the MIV of Hernán and Robins (2006), except that the average is taken over both $A = 1$ and $A = 0$ groups.

- $\widehat{\tau}_{\mathsf{ATE}} = 9432.608, \quad \mathsf{SE} = 1434.856, \quad 95\%\ \mathsf{CI} = (7064.505, 12191.134)$

Outline

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2. Review: Linear Regression
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Endogenous vs Exogenous Regressors

- Consider a linear model:

$$Y = \beta_0 + \beta_A^\intercal A + \beta_X^\intercal X + \epsilon \ , \quad \mathrm{E}(\epsilon|X) = 0 \tag{3}$$

- We allow multiple, continuous treatments

Definition (Endogeneity and Exogeneity)

Given the linear model (3)

- If $\mathrm{E}(\epsilon|A, X) \neq 0$, A is **endogenous**
- If $\mathrm{E}(\epsilon|A, X) = 0$, A is **exogenous**

- **Terminology:** Endogeneity $\Rightarrow \mathrm{E}(\epsilon|A, X) \neq 0$; Exogeneity $\Rightarrow \mathrm{E}(\epsilon|A, X) = 0$

Linear IV Model

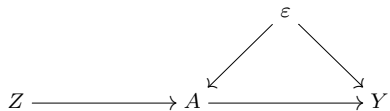
- If A is endogenous, the OLS estimator is inconsistent
- We need additional information to estimate β **consistently**

- **Solution:** Use an IV Z

Definition (Linear IV Model)

$$Y = \beta_0 + \beta_A^\intercal A + \beta_X^\intercal X + \epsilon \quad \text{with} \quad E(\epsilon|X) = E(\epsilon|Z, X) = 0$$

- Allows $E(\epsilon|A, X) \neq 0$
- Requires Z uncorrelated with error ε
- **Relevance**: Z must be correlated with A to be **useful**



Just-Identified Case

Definition (Linear IV Model)

$$Y = \beta_0 + \beta_A^\intercal A + \epsilon \quad \text{with} \quad E(\epsilon) = E(\epsilon|Z) = 0$$

- Assume (i) $\dim(Z) = \dim(A)$; (ii) $X = \varnothing$; (iii) (Relevance): $\mathrm{Cov}(Z, A)$ is invertible

- Then, we have

$$\begin{aligned} & \mathrm{Cov}(Z, Y - \beta_0 - \beta_A^\intercal A) = \mathrm{Cov}(Z, \epsilon) = 0 \\ \Rightarrow \quad & \mathrm{Cov}(Z, Y) = \mathrm{Cov}(Z, \beta_0) + \mathrm{Cov}(Z, \beta_A^\intercal A) = \mathrm{Cov}(Z, A)\beta_A \\ \Rightarrow \quad & \beta_A = \mathrm{Cov}(Z, A)^{-1}\mathrm{Cov}(Z, Y) \end{aligned}$$

- The corresponding **IV estimator** is:

$$\widehat{\beta}_{\mathsf{IV}} = \widehat{\mathrm{Cov}}(Z, A)^{-1}\widehat{\mathrm{Cov}}(Z, Y)$$

Example: Scalar IV

- Model:

$$Y = \beta_0 + \beta_A A + \epsilon, \quad E(\epsilon) = 0, \ \mathrm{E}(\epsilon|Z) = 0$$

- Then:

$$\beta_A = \frac{\mathrm{Cov}(Z, Y)}{\mathrm{Cov}(Z, A)}$$

- If Z is binary:

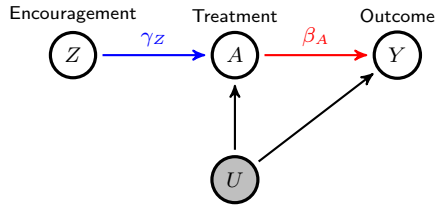
$$\beta_A = \frac{E(Y \mid Z = 1) - E(Y \mid Z = 0)}{E(A \mid Z = 1) - E(A \mid Z = 0)}$$

- Reduces to the Wald statistic!

Structural Model

- Consider

$$A = \gamma_0 + \gamma_Z Z + \eta \qquad \mathrm{E}(\eta|Z) = 0$$
$$Y = \beta_0 + \beta_A A + \epsilon \qquad \mathrm{E}(\epsilon|Z) = 0$$



- This implies

$$\begin{aligned} Y &= \beta_0 + \beta_A A + \epsilon \\ &= \beta_0 + \beta_A(\gamma_0 + \gamma_Z Z + \eta) + \epsilon \\ &= (\beta_0 + \beta_A\gamma_0) + (\beta_A\gamma_Z)Z + (\beta_A\eta + \epsilon) \end{aligned}$$

- This implies

$$\left.\begin{aligned} \mathrm{Cov}(Z,Y) &= \beta_A\gamma_Z \\ \mathrm{Cov}(Z,A) &= \gamma_Z \end{aligned}\right\} \quad \Rightarrow \quad \beta_A = \frac{\mathrm{Cov}(Z,Y)}{\mathrm{Cov}(Z,A)}$$

Over-Identified Case

- $d_Z = \dim(Z) \geqslant \dim(A) = d_A$

- More moment conditions than parameters

$$\mathrm{Cov}(Z, Y - \beta_0 - \beta_A^\intercal A) = \mathrm{Cov}(Z, \epsilon) = 0 \in \mathbb{R}^{d_Z}$$

$$\Rightarrow \quad \mathrm{Cov}(Z, Y) = \mathrm{Cov}(Z, \beta_0) + \mathrm{Cov}(Z, \beta_A^\intercal A) = \mathrm{Cov}(Z, A)\beta_A \in \mathbb{R}^{d_Z} \qquad \text{whereas} \qquad \beta_A \in \mathbb{R}^{d_A}$$

- Sample moment equations may not have exact solution

- **Solution: Two-Stage Least Squares (TSLS)**

Two-Stage Least Squares (TSLS, 2SLS)

Step 1: First stage regression:

$$\widehat{A} \leftarrow \texttt{lm}(A \sim 1 + Z + X)$$

Step 2: Second stage regression:

$$\widehat{\beta}_{\text{tsls}} = \text{coefficient of } \widehat{A} \text{ in } \texttt{lm}(Y \sim 1 + \widehat{A} + X)$$

- Closed-form representation:

$$\widehat{\beta}_{\text{tsls}} = \text{2nd component of } \left(\widehat{\mathcal{D}}^\intercal \widehat{\mathcal{D}}\right)^{-1} \left(\widehat{\mathcal{D}}^\intercal \mathcal{Y}\right) \quad \text{where} \quad \widehat{\mathcal{D}} = \begin{bmatrix} 1 & \widehat{A}_1 & X_1^\intercal \\ 1 & \widehat{A}_2 & X_2^\intercal \\ & \vdots & \\ 1 & \widehat{A}_N & X_N^\intercal \end{bmatrix} \in \mathbb{R}^{N \times (2 + \dim(X))}, \; \mathcal{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_N \end{bmatrix} \in \mathbb{R}^{N \times 1}$$

TSLS: Why It Works

- Let $\mathcal{D} = \begin{bmatrix} 1 & A_1 & X_1^\intercal \\ & \vdots & \\ 1 & A_N & X_N^\intercal \end{bmatrix}$

- Substitute true model:

$$\begin{aligned}
(\hat{\mathcal{D}}^\intercal \hat{\mathcal{D}})^{-1}(\hat{\mathcal{D}}^\intercal \mathcal{Y}) &= (\hat{\mathcal{D}}^\intercal \hat{\mathcal{D}})^{-1}\{\hat{\mathcal{D}}^\intercal (\mathcal{D}\beta + \vec{\epsilon})\}) \\
&= (\hat{\mathcal{D}}^\intercal \hat{\mathcal{D}})^{-1}(\hat{\mathcal{D}}^\intercal \mathcal{D}\beta) + (\hat{\mathcal{D}}^\intercal \hat{\mathcal{D}})^{-1}(\hat{\mathcal{D}}^\intercal \vec{\epsilon}) \\
&\overset{\star}{=} (\hat{\mathcal{D}}^\intercal \hat{\mathcal{D}})^{-1}(\hat{\mathcal{D}}^\intercal \hat{\mathcal{D}}\beta) + (\hat{\mathcal{D}}^\intercal \hat{\mathcal{D}})^{-1}(\hat{\mathcal{D}}^\intercal \vec{\epsilon}) \\
&= \beta + \left(\frac{1}{N}\hat{\mathcal{D}}^\intercal \hat{\mathcal{D}}\right)^{-1}\left(\frac{1}{N}\hat{\mathcal{D}}^\intercal \vec{\epsilon}\right) \\
&\xrightarrow{P} \beta
\end{aligned}$$

- $\star$ = the first stage OLS $\texttt{lm}(A \sim 1 + Z + X)$ implies that $\hat{\mathcal{D}}^\intercal \mathcal{D} = \hat{\mathcal{D}}^\intercal \hat{\mathcal{D}}$

- Suffices to show the last line

TSLS: Why It Works

- In addition:

$$\frac{1}{N}\widehat{\mathcal{D}}^{\intercal}\vec{\epsilon} = \frac{1}{N}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ \widehat{A}_1 & \widehat{A}_2 & \cdots & \widehat{A}_N \\ X_1 & X_2 & \cdots & X_N \end{bmatrix}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_N \end{bmatrix} = \begin{bmatrix} \mathbb{P}(\epsilon) \\ \mathbb{P}(\widehat{A}\epsilon) \\ \mathbb{P}(X\epsilon) \end{bmatrix} \xrightarrow{P} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

because of the law of large numbers:

$$\mathrm{E}(\epsilon) = \mathrm{E}(\epsilon|X) = 0$$

and

$$\begin{aligned} \mathrm{E}(\widehat{A}_i\epsilon_i|Z_1,\ldots,Z_N,X_1,\ldots,X_N) &= \mathrm{E}\big\{\big(\widehat{\gamma}_0 + \widehat{\gamma}_Z Z_i + \widehat{\gamma}_X X_i\big)\epsilon|Z_1,\ldots,Z_N,X_1,\ldots,X_N\big\} \\ &= \big(\widehat{\gamma}_0 + \widehat{\gamma}_Z Z_i + \widehat{\gamma}_X X_i\big) \times \mathrm{E}\big(\epsilon|Z_1,\ldots,Z_N,X_1,\ldots,X_N\big) = 0 \end{aligned}$$

TSLS: Variance

- Naive approach: get the SE from the second stage only, i.e., use `summary(lm(`$Y \sim 1 + \widehat{A} + X$`))`
- This approach yields an incorrect SE because it fails to account for the estimation error from the first stage
- in particular, $\widehat{A}$ is an estimated regressor, but is not incorporated in the naive approach
- Need to use the M-estimation theory; refer to the IPW lecture notes for the details

$(\widehat{\gamma}, \widehat{\beta})$ solves $\mathbb{P}\{\Psi(\gamma, \beta)\} = 0$ where $\Psi(\gamma, \beta) = \begin{bmatrix} \begin{bmatrix} 1 \\ Z \\ X \end{bmatrix} (A - \gamma_0 - \gamma_Z^\intercal Z - \gamma_X^\intercal X) \\ \begin{bmatrix} 1 \\ \gamma_0 + \gamma_Z^\intercal Z + \gamma_X^\intercal X \\ X \end{bmatrix} \{Y - \beta_0 - \beta_A^\intercal(\gamma_0 + \gamma_Z^\intercal Z + \gamma_X^\intercal X) - \beta_X^\intercal X\} \end{bmatrix}$

- Note that $\widehat{A} = \gamma_0 + \gamma_Z^\intercal Z + \gamma_X^\intercal X$

Example: 401(k)

```
## Linear 2SLS

Fit.A <- lm(p401~e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown, data=data)
data$Ahat <- predict(Fit.A)
Fit.Y <- lm(net_tfa~Ahat+age+inc+fsize+educ+db+marr+twoearn+pira+hown, data=data)
coef(summary(Fit.Y))[1:2,]

## IVReg

library(ivreg)
## Y ~ A+X|Z+X
IVREG <- ivreg(net_tfa~p401+age+inc+fsize+educ+db+marr+twoearn+pira+hown|
                 e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown,
               data=data)
coef(summary(IVREG))[1:2,]
```

- Self-TSLS $\quad \widehat{\tau}_{\text{tsls}} = 8502.323, \quad \text{SE} = 1802.521, \quad 95\%\ \text{CI} = (4969.015, 12035.630)$
- IVReg $\quad \widehat{\tau}_{\text{tsls}} = 8502.323, \quad \text{SE} = 1798.288, \quad 95\%\ \text{CI} = (4977.313, 12027.333)$

- The SEs are similar, but only by coincidence

Toy Example

```
## Toy example
N <- 1000
Z <- rnorm(N); U <- rnorm(N); A <- 2*Z + 3*U + rnorm(N); Y <- 1*A - 3*U + rnorm(N)
data.toy <- data.frame(Y=Y,A=A,Z=Z)
lm(Y~1+A)
Fit.A <- lm(A~Z, data=data.toy)
data.toy$Ahat <- predict(Fit.A)
Fit.Y <- lm(Y~Ahat, data=data.toy)
summary(Fit.Y)

IVREG <- ivreg(Y~A|Z,data=data.toy)
summary(IVREG)
```

Self-TSLS

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.08085    0.04574   1.767   0.0775 .
Ahat         0.99805    0.02157  46.263   <2e-16 ***
```

IVReg

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.08085    0.10081   0.802    0.423
A            0.99805    0.04754  20.992   <2e-16 ***
```

Outline

1. Overview
2. Review: Linear Regression
3. Instrumental Variable
4. LATE/CACE
5. ATT
6. ATE
7. TSLS
8. Nonparametric IV
9. Additional Topics
10. Assignment

Nonparametric Regression

Definition (Linear Model)

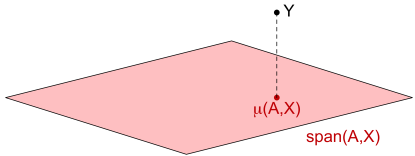
$$Y = \beta_0 + \beta_A^\intercal A + \beta_X^\intercal X + \epsilon \quad \text{with} \quad E(\epsilon|A, X) = 0$$

- Estimation of β: OLS, WLS, M-estimation, etc

Definition (Nonparametric Model)

$$Y = \mu(A, X) + \epsilon \quad \text{with} \quad E(\epsilon|A, X) = 0$$

- $\mu(A, X) = \mathrm{E}(Y|A, X)$
  - μ = projection of Y onto (A, X)
  - μ always exists



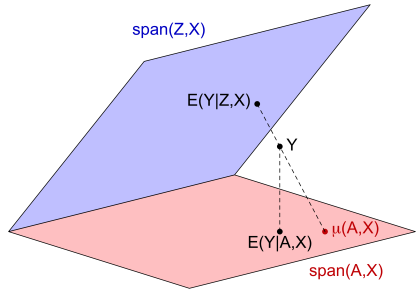
- Estimation of μ: Kernel regression (locally constant/linear), Sieve, Spline, ML, etc

Nonparametric IV

Definition (Nonparametric IV Model)

$$Y = \mu(A, X) + \epsilon \quad \text{with} \quad E(\epsilon|Z, X) = 0 \text{ but } E(\epsilon|A, X) \neq 0$$

- Equivalent form: $\mathrm{E}(Y|Z, X) = \mathrm{E}[\mu(A, X)|Z, X]$



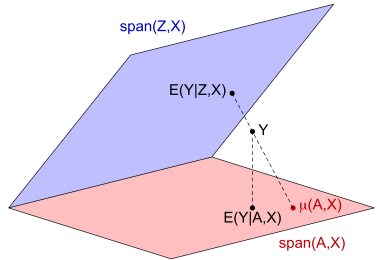
- The proj. of $\mu(A, X)$ onto (Z, X) equals $E(Y|Z, X)$ i.e., the proj. of $\mu(A, X)$ and Y onto (Z, X) coincide
- Note that $\mu(A, X) \neq E(Y|A, X)$

Nonparametric IV

Definition (Nonparametric IV Model)

$$Y = \mu(A, X) + \epsilon \quad \text{with} \quad E(\epsilon|Z, X) = 0 \text{ but } E(\epsilon|A, X) \neq 0$$

$$\Leftrightarrow \quad \mathrm{E}(Y|Z, X) = \mathrm{E}[\mu(A, X)|Z, X]$$



- **Inverse problem**: $\mu(A, X)$ should understood as a reverse regression:
  - Regression: Find a ft by projecting a known ft onto a space
  - Reverse regression: Find a ft of which projection onto a space is equal to a known ft

- An instance of a Fredholm integral equation of the first kind (1st FIE):

  Find ft f s.t. $\quad g(y) = \int K(x, y) f(x)\, dx \quad$ where K and g are given

  Find ft μ s.t. $\quad \mathrm{E}(Y|Z, X) = \int f(a|Z, X)\mu(a, X)\, da$

- **Unfortunate fact**: the solution to a 1st FIE may not exist, i.e., $\mu(A, X)$ may not be learnable
  - If $A \perp\!\!\!\perp Z|X$, μ does not exist

Identification of the ATE

Definition (Nonparametric IV Model)

$$Y = \mu(A, X) + \epsilon \quad \text{with} \quad E(\epsilon|Z, X) = 0 \text{ but } E(\epsilon|A, X) \neq 0$$
$$\Leftrightarrow \quad \mathrm{E}(Y|Z, X) = \mathrm{E}[\mu(A, X)|Z, X]$$

- Under the structural equation model, this implies $Y(a) = \mu(a, X) + \epsilon$ and $\mathrm{E}\{Y(a)|Z, X\} = \mathrm{E}[\mu(a, X)|Z, X]$

- If μ satisfying the NPIV model exists,

$$\begin{aligned}\mathrm{E}\{Y(a=1) - Y(a=0)\} &= \mathrm{E}\big[\mathrm{E}\{Y(a=1) - Y(a=0) \,\big|\, Z, X\}\big] \\ &= \mathrm{E}\big[\mathrm{E}\{\mu(1, X) - \mu(0, X) \,\big|\, Z, X\}\big] \\ &= \mathrm{E}\{\mu(1, X) - \mu(0, X)\}\end{aligned}$$

Estimation of the NPIV Function

- Note that μ is not the standard regression, so the usual ML methods cannot be used
- Need to leverage the moment condition: $\mathrm{E}[Y - \mu(A, X)|Z, X] = 0$
- This suggests

$$\text{Find } \widehat{\mu} \text{ such that } \quad 0 = \frac{1}{N}\sum_{i=1}^{N} \phi(Z_i, X_i)\{Y - \widehat{\mu}(A_i, X_i)\}$$

  where ϕ and μ are functions defined over a large function space

- Ai and Chen (2003); Newey and Powell (2003); Chen and Christensen (2018) used sieve estimation; approximating function spaces with basis functions
- Singh et al. (2019); Mastouri et al. (2021); Ghassami et al. (2022) used reproducing kernel Hilbert spaces

Example: 401(k)

- Sieve estimation typically involves increasing the degree of the basis functions as N grows
- In other words, a data-driven justification would normally be needed for using quadratic terms in the analysis below
- Here, however, we fix the degree to quadratic terms to illustrate the nonlinear IV model

```
library(ivreg)

IVReg.Quad <- function(data){
  IVREG <- ivreg(net_tfa~(p401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2|
                   (e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)^2,
                 data=data)
  data1 <- data0 <- data; data1$p401 <- 1; data0$p401 <- 0
  return( mean(predict(IVREG,data1)-predict(IVREG,data0)) )
}
Effect <- IVReg.Quad(data)
N <- nrow(data)
IVReg.Quad.Boot <- sapply(1:100,function(ii){IVReg.Quad(data[sample(N,N,replace=TRUE),])})
c(Effect, sd(IVReg.Quad.Boot), quantile(IVReg.Quad.Boot,c(0.025,0.975)))
```

- Nonlinear IV Model: $\widehat{\tau}_{\text{tsls}} = 9636.083, \quad \text{SE} = 1295.114, \quad 95\%\ \text{CI} = (7172.963, 11746.674)$

Outline

1. Overview
2. Review: Linear Regression
3. Instrumental Variable
4. LATE/CACE
5. ATT
6. ATE
7. TSLS
8. Nonparametric IV
9. Additional Topics
10. Assignment

Weak IV

- Consider

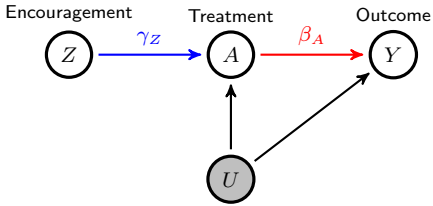
$$A = \gamma_0 + \gamma_Z Z + \eta \qquad \mathrm{E}(\eta|Z) = 0$$
$$Y = \beta_0 + \beta_A A + \epsilon \qquad \mathrm{E}(\epsilon|Z) = 0$$

- This implies

$$\left.\begin{array}{l} \mathrm{Cov}(Z,Y) = \beta_A \gamma_Z \\ \mathrm{Cov}(Z,A) = \gamma_Z \end{array}\right\} \quad \Rightarrow \quad \beta_A = \frac{\mathrm{Cov}(Z,Y)}{\mathrm{Cov}(Z,A)}$$

$$\Rightarrow \quad \widehat{\beta}_A = \frac{\widehat{\mathrm{Cov}}(Z,Y)}{\widehat{\mathrm{Cov}}(Z,A)}$$

- What if $\gamma_Z \simeq 0$?
- Relevance assumption is nearly violated
- **Weak IV**



Weak IV

- One can empirically assess the weak IV problem

- Staiger and Stock (1997): Assess the test statistic for $H_0 : \gamma_Z = 0$ in the first stage regression:

$$A = \gamma_0 + \gamma_Z Z + \eta \qquad \mathrm{E}(\eta|Z) = 0$$

$$Y = \beta_0 + \beta_A A + \epsilon \qquad \mathrm{E}(\epsilon|Z) = 0$$

  - Specifically, compare the F-statistic of the first stage regression to 10

- If $H_0 : \gamma_Z = 0$ is not rejected,
  - get better IVs (?!)
  - if you have many IVs, drop weaker IVs
  - use limited information maximum likelihood (LIML) instead of the TSLS (Anderson and Rubin, 1950)

Testing Endogeneity

- Consider the two models:

$$\text{Model 1:} \quad Y = \beta_0 + \beta_A A + \epsilon \ , \ \mathrm{E}(\epsilon|Z) = 0 \ , \ \mathrm{E}(\epsilon|A) \neq 0$$

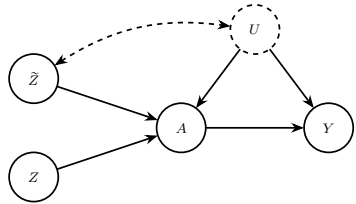
versus

$$\text{Model 2:} \quad Y = \beta_0 + \beta_A A + \epsilon \ , \ \mathrm{E}(\epsilon|A) = 0$$

- Note that $\widehat{\beta}_{\text{tsls}}$ is consistent in both models, but $\widehat{\beta}_{\text{ols}}$ is only consistent in Model 2

- Durbin-Wu-Hausman test (Durbin, 1954; Wu, 1973; Hausman, 1978): $H_0 : \text{plim}(\widehat{\beta}_{\text{tsls}}) = \text{plim}(\widehat{\beta}_{\text{ols}})$
  - If H_0 is rejected, this means that A is endogenous

Over-identification Test



- Suppose that
  - we have m IV candidates: $(Z_1, \ldots, Z_m)$;
  - we a priori know there is at least one valid IV
- The goal is to assess whether all m IV candidates are valid, specifically whether they are exogenous
- Sargan-Hansen test (aka J-test; Sargan (1958); Hansen (1982)) H_0 : all Z are exogenous
- If H_0 is rejected:
  - this means that at least one and up to $m-1$ IV candidates may be endogenous
  - also means that m IV candidates are unnecessarily over-identified

Example: 401(k)

```
IVREG <- ivreg(net_tfa~(p401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)|
                (e401+age+inc+fsize+educ+db+marr+twoearn+pira+hown),
               data=data)
summary(IVREG)
```

```
Diagnostic tests:
                  df1  df2 statistic p-value
Weak instruments    1 9904 12594.964 < 2e-16 ***
Wu-Hausman          1 9903     6.752 0.00938 **
Sargan              0   NA        NA      NA
```

Example: 401(k)

- Use $\widetilde{Z}$ that is associated with the residual $\epsilon = Y - \widehat{\beta}_{\text{ssls}} A$

```
data$residual <- IVREG$residuals
IVREG.Invalid <-
  ivreg(net_tfa~(p401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)|
          (e401+residual+age+inc+fsize+educ+db+marr+twoearn+pira+hown),
        data=data)
summary(IVREG.Invalid)
```

```
Diagnostic tests:
                 df1  df2 statistic p-value
Weak instruments   2 9903  6310.546  <2e-16 ***
Wu-Hausman         1 9903     4.165  0.0413 *
Sargan             1   NA  9907.564  <2e-16 ***
```

- Make $\widetilde{Z}$ that is independent with the residual $\epsilon = Y - \widehat{\beta}_{\text{ssls}} A$

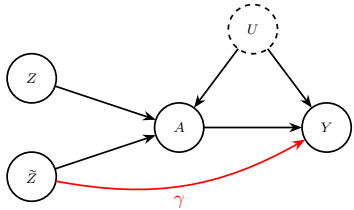
```
data$noise <- rbinom(nrow(data),1,0.5)
IVREG.Noise <-
  ivreg(net_tfa~(p401+age+inc+fsize+educ+db+marr+twoearn+pira+hown)|
          (e401+noise+age+inc+fsize+educ+db+marr+twoearn+pira+hown),
        data=data)
summary(IVREG.Noise)
```

```
Diagnostic tests:
                 df1  df2 statistic p-value
Weak instruments   2 9903  6298.318 < 2e-16 ***
Wu-Hausman         1 9903     6.754 0.00937 **
Sargan             1   NA     0.000 0.99724
```

Violation Against Exclusion Restriction

- Suppose that
  - we have m IV candidates: $Z = (Z_1, \ldots, Z_m)$;
  - allow for the violation of the exclusion restriction

$$Y = \beta_0 + \beta_A A + \sum_{j=1}^{m} \gamma_j Z_j + \epsilon \ , \ \mathrm{E}(\epsilon|Z) = 0 \ , \ \mathrm{E}(\epsilon|A) \neq 0$$



- $\gamma \neq 0$: violation of the exclusion restriction

- Let q = number of valid IVs; $m - q$ = number of invalid IVs

- Kang et al. (2016)'s contribution:
  - (Majority rule): If $q \geqslant m/2$, then β_A can be identified;
  - (Estimator): Some invalid some valid IV estimator (sisVIVE); use ℓ_1 penalty to select valid IVs
    Established consistency of the sisVIVE

Outline

1. Overview
2. Review: Linear Regression
3. Instrumental Variable
4. LATE/CACE
5. ATT
6. ATE
7. TSLS
8. Nonparametric IV
9. Additional Topics
10. Assignment

Assignment

- Analyze the dataset `causalweight::JC` (Job Corps) with the following variables:
  - $X =$ (`female, age, white, black, hispanic`)
  - $Z =$ `assignment`
  - $A = \mathbb{1}(\texttt{train1} + \texttt{train2} > 0) =$ participated in vocational training in year 1 or 2
  - $Y =$ `earny4`
- Goal: Estimate the causal effect of vocational training participation on 4th-year weekly earnings, using assignment as an IV
- Apply the IV methods as outlined in the following steps:
  - First, explain why an IV method would be useful and discuss whether Z appears to be a valid IV
  - Next, estimate the LATE, ATT, ATE, and the treatment effect assuming the linear IV model
  - For each method, indicate which of the four IV assumptions is used
  - Interpret the diagnostic tests in the `ivreg` object
- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

References

Ai, C. and Chen, X. (2003). Efficient estimation of models with conditional moment restrictions containing unknown functions. *Econometrica*, 71(6):1795–1843.

Anderson, T. W. and Rubin, H. (1950). The asymptotic properties of estimates of the parameters of a single equation in a complete system of stochastic equations. *The Annals of Mathematical Statistics*, pages 570–582.

Angrist, J. D. (1990). Lifetime earnings and the Vietnam era draft lottery: evidence from social security administrative records. *American Economic Review*, 80:313–336.

Angrist, J. D. and Evans, W. N. (1998). Children and their parents' labor supply: Evidence from exogenous variation in family size. *American Economic Review*, 88:450–477.

Angrist, J. D., Imbens, G. W., and Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *Journal of the American Statistical Association*, 91(434):444–455.

Angrist, J. D. and Krueger, A. B. (1991). Does compulsory school attendance affect schooling and earnings? *Quarterly Journal of Economics*, 106:979–1014.

Card, D. (1993). Using geographic variation in college proximity to estimate the return to schooling. Technical report, National Bureau of Economic Research.

Chen, C.-H., Budas, G. R., Churchill, E. N., Disatnik, M.-H., Hurley, T. D., and Mochly-Rosen, D. (2008). Activation of aldehyde dehydrogenase-2 reduces ischemic damage to the heart. *Science*, 321(5895):1493–1495.

Chen, X. and Christensen, T. M. (2018). Optimal sup-norm rates and uniform inference on nonlinear functionals of nonparametric iv regression. *Quantitative Economics*, 9(1):39–84.

Durbin, J. (1954). Errors in variables. *Review of the International Statistical Institute*, 22:23–32.

Ghassami, A., Ying, A., Shpitser, I., and Tchetgen Tchetgen, E. (2022). Minimax kernel machine learning for a class of doubly robust functionals with application to proximal causal inference. In Camps-Valls, G., Ruiz, F. J. R., and Valera, I., editors, *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*, volume 151 of *Proceedings of Machine Learning Research*, pages 7210–7239. PMLR.

Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica: Journal of the econometric society*, pages 1029–1054.

References

Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica*, 46(6):1251–1271.

Hearst, N., Newman, T. B., and Hulley, S. B. (1986). Delayed effects of the military draft on mortality. *New England Journal of Medicine*, 314:620–624.

Hernán, M. A. and Robins, J. M. (2006). Instruments for causal inference: An epidemiologist's dream? *Epidemiology*, 17(4):360–372.

Kang, H., Zhang, A., Cai, T. T., and Small, D. S. (2016). Instrumental variables estimation with some invalid instruments and its application to mendelian randomization. *Journal of the American Statistical Association*, 111(513):132–144.

Kitagawa, T. (2015). A test for instrument validity. *Econometrica*, 83(5):2043–2063.

Levis, A. W., Kennedy, E. H., and Keele, L. (2024). Nonparametric identification and efficient estimation of causal effects with instrumental variables. *Preprint arXiv:2402.09332*.

Liu, J., Park, C., Lee, Y., Zhang, Y., Yu, M., Robins, J. M., and Tchetgen Tchetgen, E. J. (2025). The multiplicative instrumental variable model. *Preprint arXiv:2507.09302*.

Mastouri, A., Zhu, Y., Gultchin, L., Korba, A., Silva, R., Kusner, M., Gretton, A., and Muandet, K. (2021). Proximal causal learning with kernels: Two-stage estimation and moment restriction. In Meila, M. and Zhang, T., editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 7512–7523. PMLR.

Newey, W. K. and Powell, J. L. (2003). Instrumental variable estimation of nonparametric models. *Econometrica*, 71(5):1565–1578.

Robins, J. M. (1994). Correcting for non-compliance in randomized trials using structural nested mean models. *Communications in Statistics - Theory and Methods*, 23(8):2379–2412.

Sargan, J. D. (1958). The estimation of economic relationships using instrumental variables. *Econometrica*, 26(3):393–415.

Singh, R., Sahani, M., and Gretton, A. (2019). Kernel instrumental variable regression. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

Staiger, D. and Stock, J. H. (1997). Instrumental variables regression with weak instruments. *Econometrica*, 65(3):557–586.

References

Voight, B. F., Peloso, G. M., Orho-Melander, M., Frikke-Schmidt, R., Barbalic, M., Jensen, M. K., Hindy, G., Hólm, H., Ding, E. L., and Johnson, T. (2012). Plasma HDL cholesterol and risk of myocardial infarction: a Mendelian randomisation study. *The Lancet*, 380:572–580.

Wang, L. and Tchetgen Tchetgen, E. J. (2018). Bounded, efficient and multiply robust estimation of average treatment effects using instrumental variables. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 80(3):531–550.

Wu, D.-M. (1973). Alternative tests of independence between stochastic regressors and disturbances. *Econometrica*, 41(4):733–750.